

# Vedant Jha

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## Education

**Manipal Institute of Technology, Manipal**  
*B.Tech. in Information Technology; CGPA: 7.58*

*Manipal, India*  
2024 – 2028

Coursework: Data Structures, Design and Analysis of Algorithms, Operating Systems, Object Oriented Software Development Lab (Java), Database Management Systems (SQL), Data Communication and Computer Networks, Digital Systems and Verilog, Object Oriented Programming (Java), Data Analysis and Visualization, Discrete Mathematics, Probability and Optimization

## Skills

**Programming Languages:** Python, C, Java, SQL, MATLAB, Verilog, HTML

**Machine Learning:** Supervised Learning, Unsupervised Learning, Regression, Classification, Clustering, Ensemble Methods (Random Forest, XGBoost), Feature Engineering, Model Evaluation, Scikit-learn

**Deep Learning & Computer Vision:** PyTorch, TensorFlow, Neural Networks, Object Detection (YOLOv8), Image Classification, OpenCV

**Generative AI, LLMs & Agents:** Large Language Models (LLMs), Prompt Engineering, Multi-Agent Systems, Agentic Workflows, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG) (Basics)

**Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn, Exploratory Data Analysis (EDA)

**Tools, Frameworks & Platforms:** FastAPI, Streamlit, MySQL, JavaFX, Maven, LTSpice, Cisco Packet Tracer, Git

## Internship Experience

**Capgemini**

*Associate Digital Analyst*

*Aeroli Campus, India*  
25 May – 30 June 2026

- Worked on a multi-agent AI project, Email Intelligent Agent, built using an MCP (Model Context Protocol) server for tool exposure and a separate orchestrator agent that interpreted the prompt/context at runtime to decide which tools needed to be executed, decoupling business logic from control flow
- Designed the orchestrator to reason over available tools at runtime and chain them (fetch, then classify, then summarize, then export) rather than following a hardcoded pipeline, making the system extensible to new tools without changing orchestration logic
- Engineered a checkpointing mechanism that persisted the timestamp of the last successfully processed email, enabling idempotent, incremental mailbox syncs and preventing duplicate processing on subsequent runs
- Designed LLM prompts to jointly parse conversation threads, infer sender intent, and output a structured criticality score with an escalation flag, effectively turning unstructured email content into a machine-actionable signal
- Generated a recommended next action per email thread by having the LLM reason over conversation summary and criticality together, reducing manual triage effort for the reviewing team
- Built a structured export pipeline writing results to both JSON (for programmatic downstream use) and Excel (sender, recipient, thread, summary, recommended action, direct mail link) for non-technical stakeholders to review

## Positions of Responsibility

**Management Committee Member – The Consulting Club at Manipal (TCCAM)**  
*Student Club, MIT Manipal*

*Manipal, India*  
2024 – Present

- Conducted a comprehensive case study on EV battery technologies and market dynamics for an organization, analyzing market trends, customer requirements, and sales potential
- Prepared a detailed report providing strategic insights and business support recommendations
- Authored an analytical article on Singapore's economic transformation from a developing nation to a global financial hub

**Working Committee Member – Apple Developers Group**  
*Student Club, MIT Manipal*

*Manipal, India*  
Nov 2024 – Mar 2026

- Conducted workshops on Machine Learning fundamentals, covering supervised and unsupervised models such as Linear Regression, Decision Trees, and K-Means clustering

**Management Committee Member – Finova**  
*Student Club, MIT Manipal*

*Manipal, India*  
Nov 2024 – Present

- Studied key factors affecting stock market movements, including macroeconomic indicators, interest rates, and investor sentiment

## Projects

**Hotel Management System**

*JavaFX, Maven, Scene Builder, CSS*

- Developed a full-featured Hotel Management System using JavaFX, Maven, and Scene Builder with CSS-based UI styling that handles room booking, customer management, billing, and reservation tracking
- Designed an interactive GUI with modular architecture and efficient data handling

## **Car Dataset Analysis and Visualization**

*Python, NumPy, Matplotlib, Seaborn*

- Performed exploratory data analysis on automobile datasets using NumPy, Matplotlib, and Seaborn that derives insights on pricing, features, and performance trends

## **WealthWise: AI-Powered Financial Assistance**

*Streamlit, Ollama, NSE/Nifty 50 API*

- Developed a Streamlit web app integrating an Ollama-based chatbot and live NSE/Nifty 50 APIs, with user authentication, that delivers personalized financial insights and data tracking

## **Pothole Detection System (Ongoing)**

*YOLOv8, Computer Vision*

- Developing a pothole detection system using YOLOv8 that detects and classifies road damage from images/videos for enhanced road safety and analysis

## **Stock Price Prediction using Machine Learning**

*Python, Random Forest, XGBoost*

- Developed a machine learning model using 10 years of historical market data and ensemble methods (Random Forest, XGBoost) that predicts stock prices via regression